



Structural Studies of Fourth-Generation EGFR Inhibitors Reveal Insights into Selective T790M and C797S Targeting

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Abstract

Inhibitors targeting mutant EGFR remain a persistent need in combating drug resistance in non-small cell lung cancer. To better understand the molecular factors involved in targeting T790M and C797S mutations, we determined X-ray cocrystal structures of fourth-generation inhibitors BI-8128 and BI-4732. Analysis from molecular dynamics and thermodynamic integration calculations correlated with biochemical and cellular measurements indicate that BI-8128 binds

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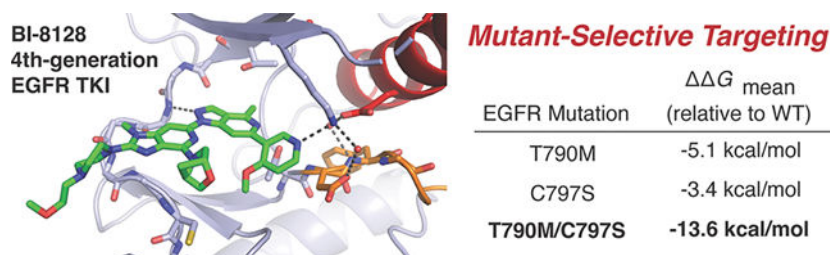
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the double T790M/C797S more strongly than the single mutations individually. This observation showcases strengths in the design of these fourth-generation EGFR inhibitors as profile criteria require drugs to inhibit an array of oncogenic and drug resistance mutations.

Graphical Abstract



Keywords

EGFR; kinase inhibitors; non-small cell lung cancer; structural biology; crystallography; molecular dynamics; targeted therapy

Activating mutations within the kinase domain of the epidermal growth factor receptor (EGFR) are often associated with oncogenic properties in non-small cell lung cancer (NSCLC) and represent biomarkers for selecting effective small-molecule therapies.^{1,2} Initially, activating EGFR mutations, such as L858R and 19del, can be treated with tyrosine kinase inhibitors (TKIs), most commonly AZD9291 (osimertinib).^{2–5} This drug is an irreversible inhibitor that targets C797 located adjacent to the ATP-substrate site and readily forms covalent bonds with drugs harboring a Michael acceptor functional group.^{6,7} Additionally, osimertinib targets both initial activating EGFR mutations as well as secondary acquired T790M “gatekeeper” mutation that renders certain TKIs ineffective.⁸ More recently, the C797-targeting irreversible TKI YH25448 (lazertinib) in combination with the dual EGFR/c-MET targeting antibody amivantimab was shown to be superior to osimertinib monotherapy.^{9–12} In both cases, NSCLC tumors eventually become resistant to osimertinib and lazertinib by acquiring the C797S mutation that prevents these drugs from forming their potency-enabling covalent bonds.^{13,14}

The discovery of drug resistance to covalent third-generation TKIs has initiated efforts to discover molecules that can inhibit EGFR containing T790M and C797S while simultaneously exhibiting limited activity against wild-type (WT) to ensure efficacy and safety. Diverse compounds have emerged, such as trisubstituted imidazoles,^{15–18} brigatinib,¹⁹ 18k,²⁰ CH7233163,²¹ BLU-945,²² and JND3229²³ as well as the unique aminobenzimidazole macrocycle BI-4020.^{24,25} Stemming from BI-4020, an advanced variant BI-4732, and related BI-8128, were developed showing impressive preclinical efficacy as well as improved blood-brain-barrier (BBB) penetration.²⁶ Available *in vitro* cellular data show meaningful nanomolar potencies against treatment naive mutations (L858R and del19) and the corresponding T790M/C797S drug-resistant analogues while being impressively inactive against WT-containing cells (Table 1). Both compounds are

structurally unique compared to their contemporaries and differ only with respect to their piperazine-derived solubilization moieties (*R*).

To understand the structural basis for how BI-8128 and BI-4732 target T790M/C797S mutant EGFR, we carried out soaking of these molecules in WT EGFR crystals. These crystals contain the EGFR kinase domain packed into repeating active conformations with the α C-helix rotated into the “in” conformation.²⁷ Diffraction data sets at 2.1 and 2.4 Å high resolution were obtained for BI-8128 and BI-4732, respectively, and indexed to unit cells and space groups (*I*₂₃) consistent with previous findings (Table S1).^{18,25,28} Expectedly, both compounds, comprising a four-heterocycle architecture (Figures 1 and 2A), are found to bind to EGFR at the ATP-binding site producing practically identical interactions (Figure 2B,C). The distinct solubilization groups extend from the imidazolopyridine forming no apparent interactions with EGFR kinase. The ligands are well modeled as evident by the correspondence to mF_o-DF_c simulated annealing omit maps (Figure 2D). These structures showcase that the equivalent activity profiles (Table 1) are associated with nearly identical binding interactions with these molecules targeting the active kinase conformation.

A closer examination of both X-ray cocrystal structures reveals a series of interactions with the EGFR kinase domain (Figure 3, Figure S1). The imidazolopyridine is anchored to the hinge M793 accepting an H-bond from the backbone amide (Figure 3A). The methyl group on the imidazolopyridine is positioned near the α C of P794 and at a distance consistent with a van der Waals contact (~3.8 Å; Figure 3A). The 4-methoxypyridine is positioned off the pyrazolopyrimidine such that the pyridine accepts an H-bond from the catalytic lysine (K745), which is consistent with fourth-generation EGFR TKIs.^{18,29} K745 is also observed to make interactions with solvent waters and a salt bridge with E762 as a hallmark of an active kinase conformation (Figure 3B). Additionally, the bicyclic morpholine group is positioned along hydrophobic region II (HR II) making van der Waals contacts with F723 that is found “tucked” beneath the P-loop (Figure 3C,D). The 4-methoxypyridine is seen within van der Waals distance with these residues as well suggesting that these groups stabilize this amino acid side chain and the P-loop β -sheet. These interactions are practically identical for the binding of BI-4732 (Figure S1) and the unique piperazine moiety is found positioned into solvent.

To better appreciate the differences in potency of BI-8128 and BI-4732, activity assays were performed to assess the relative IC₅₀ values against WT and mutant EGFR (Table 2). Both molecules exhibited IC₅₀ values in low nanomolar concentrations, with the greatest potency against L858R/T790M/C797S at ~2–3 nM with the largest degree of mutant selectivity being observed for BI-8128. These results are consistent with the observation that BI-8128 and BI-4732 are highly potent against a set of mutations, but indicate that they exhibit similar potency against WT. Interestingly, despite being most potent against L858R/T790M/C797S, BI-8128 is less potent against the double T790M/C797S and single mutant C797S showing that this compound has a unique preference for the triple mutation. Previous *in vitro* cellular data (Table 1) confirm lower potency against WT-containing cell lines while our biochemical results indicate relatively higher potency (Table 2), which we speculate may be due to differences in cellular contexts (e.g., dimerization dependence on signaling for WT and not the mutations). Additionally, the origin for the difference in mutant profiles between

the two compounds, namely, that BI-4732 is equally potent against all mutations and WT, is unclear but could be caused by uncharacterized functional consequences stemming from the piperazine-derived solubilization groups (*R* in Figure 1).

We next employed molecular dynamics (MD) simulations utilizing our X-ray cocrystal structures and thermodynamic integration (TI) calculations to quantify how T790M and C797S mutations impact the binding energetics of BI-8128. Previous studies have shown that MD simulations along with free-energy calculations including TI can accurately reproduce and rationalize structure changes in protein systems.^{30–32} Given that the L858R/T790M/C797S mutation is selectively inhibited by BI-8128, we sought to gain deeper appreciation for BI-8128 binding in the context of amino substitutions for T790M and C797S as our X-ray cocrystal structures were determined with the WT kinase. To that end, we performed thermodynamic TI which is a rigorous free energy calculation method that quantifies the difference in free energy between states, including amino acid substitutions.^{33,34} These TI calculations indicate that the single mutation T790M and C797S kinases exhibit preferential binding for BI-8128 compared to WT, as indicated by a negative $\Delta\Delta G_{\text{mean}}$ (Figure 4). Additionally, T790M appears to have a more dominant effect on binding affinity than C797S, which is consistent with the location of the molecule in direct contact with the T790M “gatekeeper” residue (Figure 2B) and biochemical IC₅₀ values (Table 2). Interestingly, TI calculations with the double mutation T790M/C797S showcase a much stronger binding $\Delta\Delta G_{\text{mean}}$ compared to the single mutations, indicating that BI-8128 binds with greater preference to the EGFR kinase when multiple mutations are present. This is consistent with the general experimental trends in biochemical potencies for BI-8128 targeting L858R/T790M/C797S, and to a lesser extent for T790M/C797S likely indicating further complexities in explaining the structural basis for EGFR inhibition by BI-8128.

To gain a deeper understanding of the binding free energy preference for BI-8128 and the T790M/C797S double mutant, we performed analysis of interactions formed during the final 50 ns of the MD simulation trajectory (Figure 5). As consistent with the TI free energies, a general increase in H-bonds between BI-8128 and EGFR are seen for T790M/C797S compared with WT as well as single mutations T790M and C797S (Figure 5A). Additionally, an average decrease in solvent waters (Figure 5B) is accompanied by an increase in the number of protein residues observed in the binding of BI-8128 with T790M/C797S (Figure 5C). Cluster analysis of the MD trajectories revealed that the T790M/C797S double mutation induces distinct structural rearrangements within the EGFR binding pocket (Figure S2). These alterations reshape the local pocket geometry and modify the conformational landscape accessible to BI-8128. As a consequence, the ligand adopts alternative, more stabilizing binding orientations, consistent with the enhanced affinity observed for the double mutant. For instance, the C797 in WT is found in an “up” conformation that sterically clashes with BI-8128 (Figure S2A) while the mutant serine in C797S is observed only in a “down” position that does not interfere with BI-8128 (Figure S2B). The T790M “gatekeeper” is found in a position that directly contacts the ligand (“front” in Figure S2C) consistent with positive interactions observed earlier with the related macrocycle BI-4020.²⁵ This conformation of the T790M methionine is coupled with other groups of the inhibitor disrupting the P-loop β -sheet, specifically the opening up of backbone amides forming new H-bonds with the bicyclic morpholine (Figure S2D). To our

knowledge, such a perturbation is not observed in experimental structural data, but kinase inhibitor influence on P-loop conformations has been observed in other systems.^{35–37} These findings are consistent with a plausible structural basis where BI-8128 binds with enhanced interactions to the mutant EGFR kinase due to the presence of both the T790M and C797S mutations.

In this study we characterize the binding modes of BI-8128 and BI-4732 as novel fourth-generation EGFR TKIs developed for the treatment of NSCLC. Their binding modes are like other fourth-generation TKIs (Figure 1) producing H-bonds to K745^{18,29} while the hinge region is unique with respect to a van der Waals contact with the P794 and placement of a bicyclic morpholine beneath the P-loop. MD simulations, TI calculations, and cluster analysis indicate a strong preference for BI-8128 to bind the dual T790M/C797S mutation over the respective single mutations consistent with the inhibitor forming a greater number of interactions with the EGFR kinase domain. This observation is somewhat complicated as acquired resistance mutations may occur on different EGFR alleles within tumor cells; however, biochemical and cellular profiles indicate that these compounds can target diverse mutations with similar potencies. The most promising advancements in the development of fourth-generation EGFR TKIs have yielded various ATP-competitive “Type I” kinase inhibitors while several alternative binding modes such as allosteric (Type III)^{38–44} and bivalent inhibitors^{45–49} are also established. It is likely that the observations made in the context of these experimental X-ray cocrystal structures of BI-8128 and BI-4732 and related calculations will be useful to medicinal chemists presently working to optimize next-generation EGFR inhibitors as novel cancer therapies.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability Statement

X-ray cocrystal structures have been deposited in the protein data bank (PDB) under accession codes: 9DM8 and 9GNP. Other data will be made available on request.

ABBREVIATIONS

BBB blood-brain barrier

del19	exon 19 deletion
EGFR	epidermal growth factor receptor
HRII	hydrophobic region II
MD	molecular dynamics
NSCLC	nonsmall cell lung cancer
TI	thermodynamic integration
TKI	tyrosine kinase inhibitor
WT	wild-type

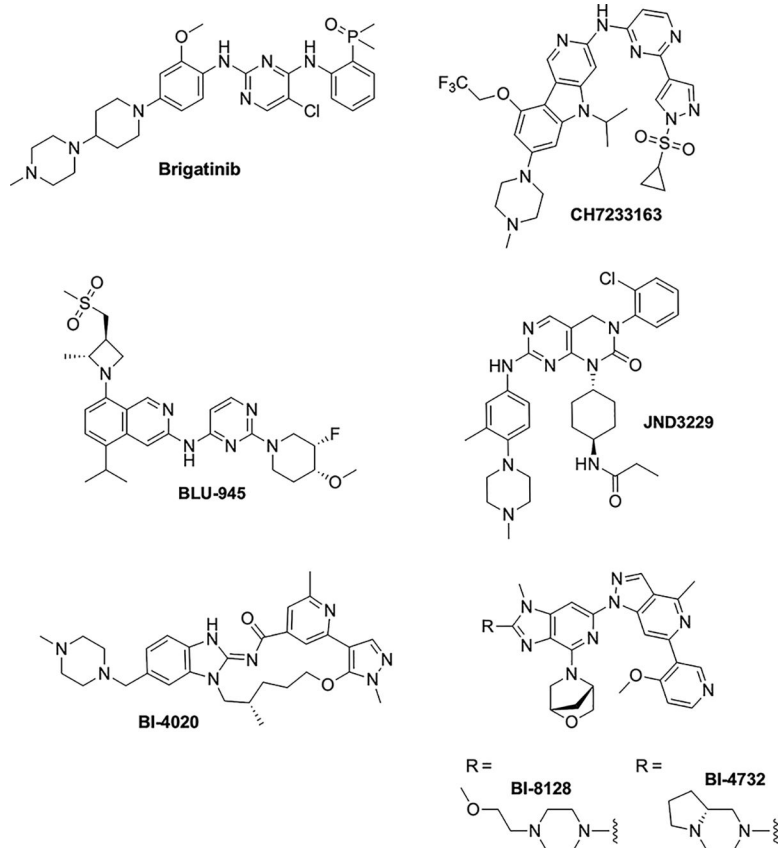
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**Figure 1.**

Representative chemical structures of fourth-generation EGFR inhibitors targeting T790M/C797S, including BI-4020, BI-4732, and BI-8128.

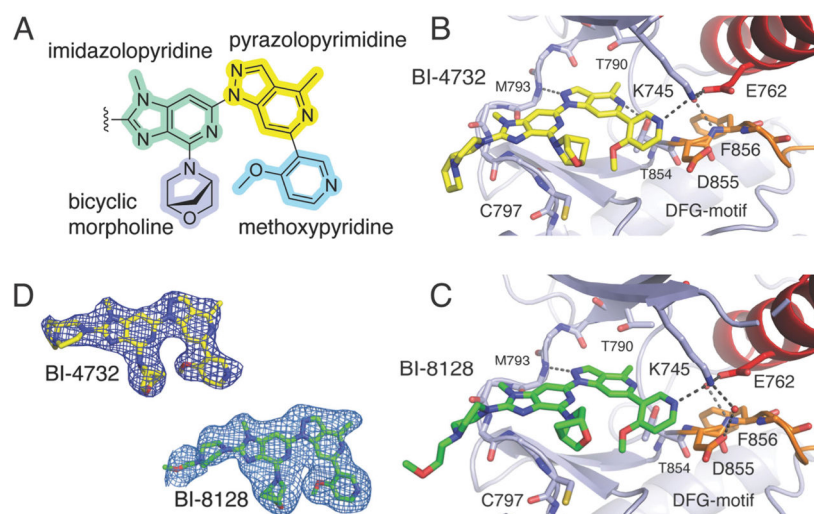


Figure 2. BI-4732 and BI-8128 in complex with the EGFR kinase domain. A) The 4-heterocycle architecture (4-methyl-2*H*-pyrazolo[4,3-*c*]pyridine “pyrazolopyrimidine”, yellow; 1-methyl-1*H*-imidazo[4,5-*c*]pyridine “imidazolopyrimidine”, teal; “4-methoxypyridine” methoxypyridine, sky blue; (1*R*,4*R*)-2-oxa-5λ²-azabicyclo[2.2.1]heptane “bicyclic morpholine”, lavender) of these fourth-generation reversible binding inhibitors. B) Overall binding modes of BI-4732 (PDB ID 9DM8) and C) BI-8128 (PDB ID 9GNP) showing the expansive and complex nature of the interactions seen in these X-ray cocrystal structures. The P-loop has been omitted for clarity despite being involved in compound binding (see Figure 3C). D) Simulated annealing mF_o-DF_c omit maps of BI-4732 and BI-8128 (contour 3σ).

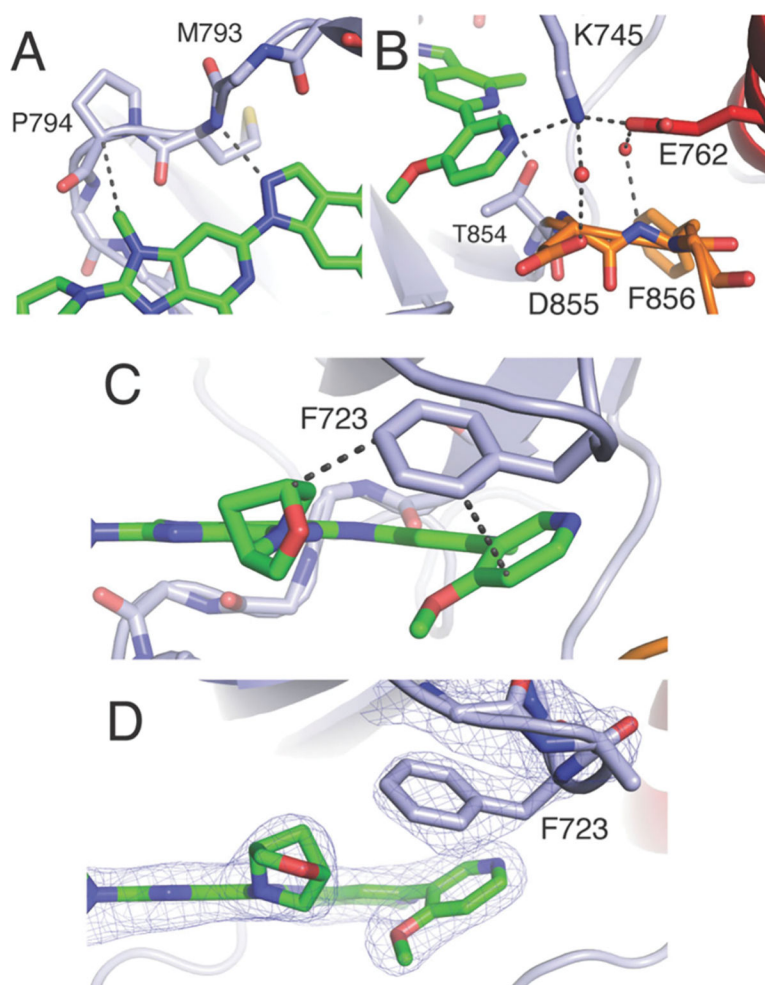


Figure 3. Diverse interactions enable binding of BI-8128 to EGFR (PDB ID 9DM8). A) Zoom around hinge-binding interactions showing H-bond accepting from M793 and van der Waals interactions with P794. B) H-bond accepting from K745 and observed interactions with the lysine ϵ -amino nitrogen. C) Hydrophobic interactions between F723 and BI-8128. D) Similar visualization as in C including positive $2mF_o-DF_c$ electron density difference map contoured to 1σ . Corresponding images for BI-4732 are visualized in Figure S1.

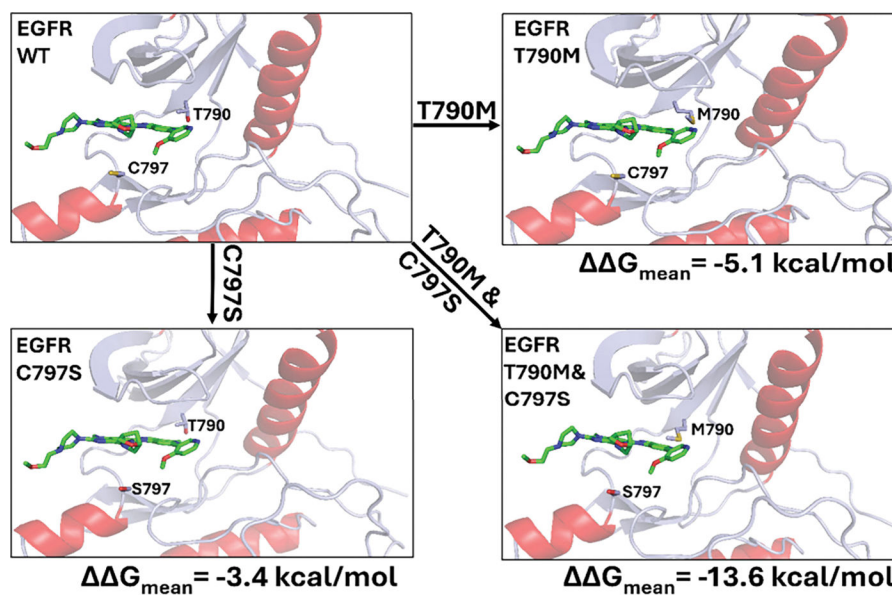
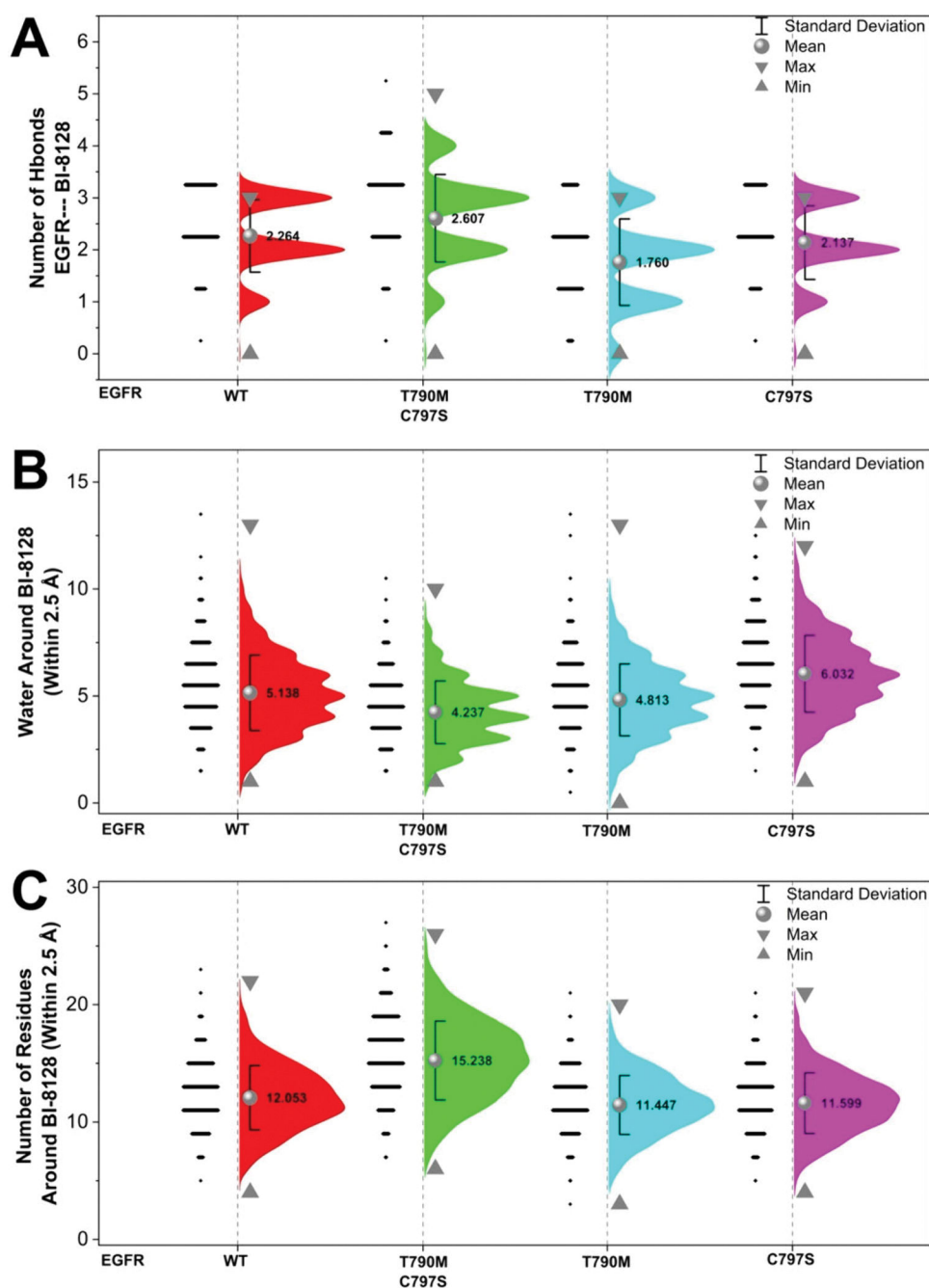


Figure 4. Free energy of binding of BI-8128 to mutant EGFR. Thermodynamic integration calculations of BI-8128 (green) binding to WT (upper left), T790M (upper right), C797S (lower left), and T790M/C797S (lower right) EGFR kinase domains.

**Figure 5.**

Assessment of interactions that complex BI-8128 in WT and mutant EGFR. A) Half-violin plots showing the distribution of H-bonds formed between the EGFR kinase and BI-8128 during the final 50 ns of each MD simulation trajectory. B) Number of water molecules within 2.5 Å of BI-8128 over the same time interval. C) Number of protein residues within 2.5 Å of BI-8128. Red, WT EGFR; green, EGFR with T790M and C797S mutations; cyan, EGFR with the T790M mutation; magenta, EGFR with the C797S mutation. Mean values

(averaged over two independent simulations) are shown as spheres; maximum and minimum values are represented by downward and upward triangles, respectively.

Table 1.*In Vitro* Cellular Data for BI-4732 and BI-8128^a

Antiproliferative Cellular Activity Values (nM)			
Cell Line	Genotype	BI-4732	BI-8128
Ba/F3	WT (added EGF)	356	318
	del19	1.1	1.7
	del19/T790M/C797S	2.6	3.3
	L858R	4.6	9.6
	L858R/T790M/C797S	7.8	16
A431	WT amplification	730	492
PC-9	parental del19 genotype	9	16
	del19/T790M/C797S	12	33

^aData made available through the opnMe initiative through Boehringer Ingelheim at <https://openme.com>. Methods from Engelhardt et al.²⁴

Table 2.
Biochemical Potencies against WT EGFR and Mutations^a

	Biochemical Potency IC ₅₀ (nM)	
	BI-8128	BI-4732
WT	12	2.7
T790M	6.7	4.2
C797S	22	5.4
T790M/C797S	10	4.9
L858R/T790M/C797S	3.0	2.3

^a Enzyme assays performed at 30 °C for 240 min at pH = 7.5 with substrate [ATP] = 1.0 mM with activity being measured with the AssayQuant Technologies Inc. sensor AQT0734 (15 μ M) and 2% DMSO in all samples. Final enzyme concentrations were 0.5 nM WT, T790M and C797S; 0.6 nM for T790M/C797S; 1.0 nM for L858R/T790M/C797S.